# CSE430:Digital Signal Processing

## LINEAR TIME INVARIANT SYSTEMS

Acknowledgement for lecture slides – Dr. Mohammad Zavid Parvez

# Linear Time Invariant Systems

A system that is both linear and time invariant is called a linear time invariant (LTI) system.

By setting the input $x[n]$ as $\delta[n]$, the impulse function, the output $h[n]$ of an LTI system is called the *impulse response* of this system.

- Time invariant: when the input is $\delta[n\text{-}k]$, the output is $h[n\text{-}k]$.
- Remember that the $x[n]$ can be represented as a linear combination of delayed impulses

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$$

# Linear Time Invariant Systems (cont.)

Hence

$$y[n] = T\left\{\sum_{k=-\infty}^{\infty} x[k]\delta[n-k]\right\} = \sum_{k=-\infty}^{\infty} x[k]T\{\delta[n-k]\}$$

$$= \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

Therefore, a LTI system is completely characterized by its impulse response $h[n]$.

# Linear Time Invariant Systems (cont.)

Convolution

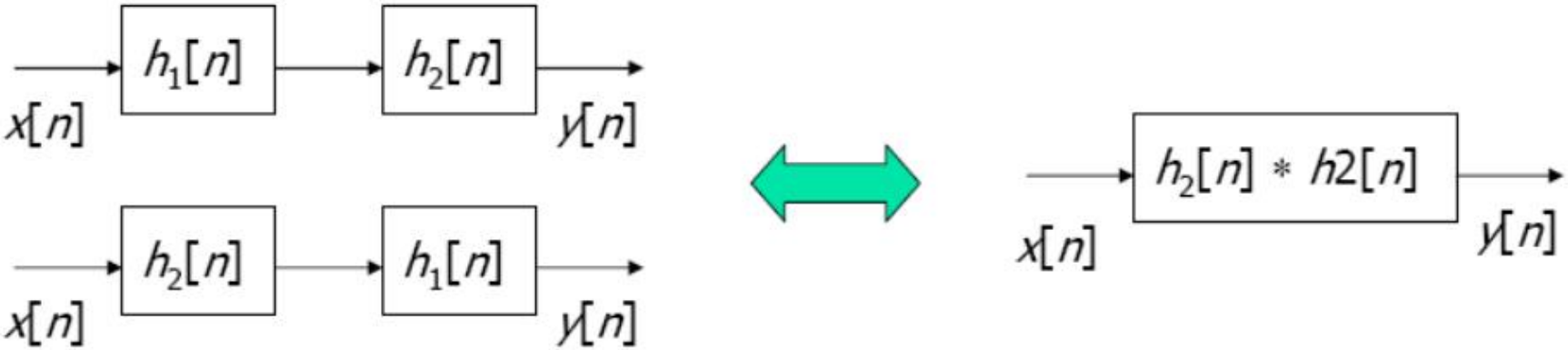
$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

- The above operation is called *convolution*, and can be written in short by $y[n] = x[n] * h[n]$.

In a LTI system, the input sample at $n = k$, represented as $x[k]\delta[n\text{-}k]$, is transformed by the system into an output sequence $x[k]h[n\text{-}k]$ for $-\infty < n < \infty$.

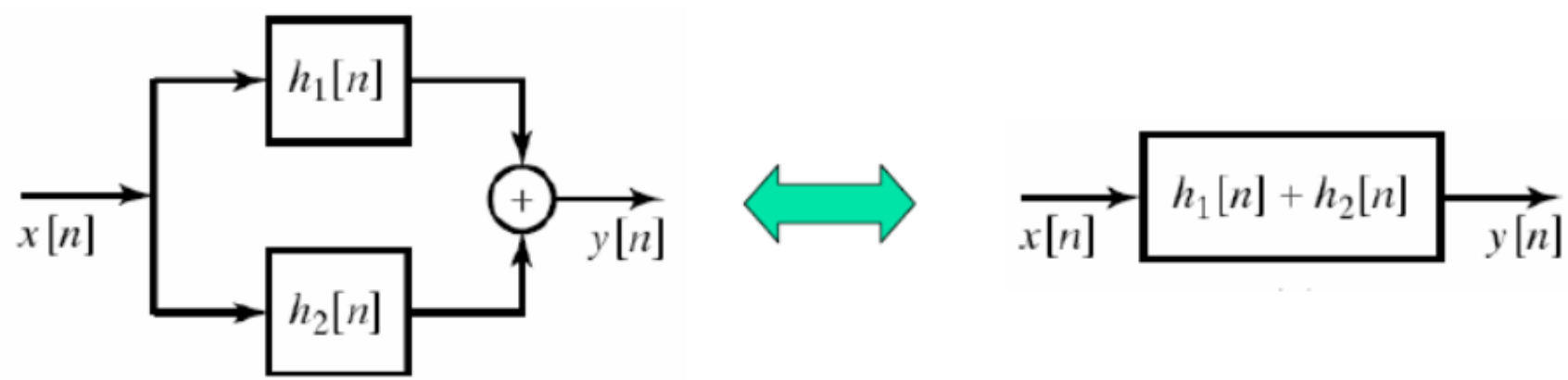
# Property of LTI System and Convolution

- Communitive
  - $x[n] * h[n] = h[n] * x[n].$
- Distributive over addition
  - $x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n].$
- Cascade connection



# Property of LTI System and Convolution (cont.)

- Parallel combination of LTI systems and its equivalent system.



# Impulse Responses of Some LTI Systems

Ideal delay: $h[n] = \delta[n-n_d]$

Moving average

$$h[n] = \begin{cases} \dfrac{1}{M_1 + M_2 + 1} & -M_1 \le n \le M_2 \\ 0 & \text{otherwise} \end{cases}$$

Accumulator

$$h[n] = \begin{cases} 1 & n \ge 0 \\ 0 & \text{otherwise} \end{cases}$$

Forward difference: $h[n] = \delta[n+1] - \delta[n]$

Backward difference: $h[n] = \delta[n] - \delta[n-1]$

Ex1. Determine the homogeneous solution of the system described by the first-order difference equation.

$$y(n) + a_1 y(n-1) = x(n) \quad \cdots (I)$$

Solution: The assumed solution obtained by setting $x(n)=0$ is

$$y_h(n) = \lambda^n$$

$$\therefore \lambda^n + a_1 \lambda^{n-1} = 0$$

$$\lambda^{n-1}(\lambda + a_1) = 0$$

$$\therefore \lambda = -a_1$$

Therefore, the solution to the homogeneous difference equ. is

$$y_h(n) = C\lambda^n = C(-a_1)^n \quad \cdots (II)$$

From (I),

$$y(n) = -a_1 y(n-1) \quad \cdots (III)$$

$$[x(n)=0]$$

$$y(0) = -a_1 y(-1)$$

From (II),

$$y_h(0) = C$$

$$\therefore C = -a_1(y-1)$$

Hence, the zero-input response of the system is

$$y_{zi}(n) = -a_1 y(-1) \cdot (-a_1)^n$$

$$= (-a_1)^{n+1} y(-1) \qquad n \geq 0$$

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Ex 2: Determine the zero-input response of the system described by the homogeneous second-order difference equation

$$y(n) - 3y(n-1) - 4y(n-2) = 0 \quad \ldots \text{(I)}$$

Soln: Firstly, we determine the solution to the homogeneous equation we assume the solution to be the exponencial,

$$y_h(n) = \lambda^n$$

Upon substitution of this solution into (I), we obtain characteristic equation.

$$\lambda^n - 3\lambda^{n-1} - 4\lambda^{n-2} = 0$$

$$\lambda^{n-2}(\lambda^2 - 3\lambda - 4) = 0$$

Therefore, roots are

$$\therefore \lambda = -1, 4$$

The general form of the solution to the homogeneous equation is

$$y_h(n) = c_1 \lambda_1^n + c_2 \lambda_2^n$$

$$= c_1 (-1)^n + c_2 (4)^n \quad \ldots \text{(II)}$$

The zero-input response of the system can be obtained from the homogeneous solution by evaluating the constants in (II), given the initial conditions $(y-1)$ and $y(-2)$

From the difference equation in (I), we have

$$y(0) = 3y(-1) + 4y(-2)$$

$$y(1) = 3y(0) + 4y(-1)$$

$$= 3[3y(-1) + 4y(-2)] + 4y(-1)$$

$$= 13y(-1) + 12y(-2)$$

From (II), we obtain

$$y(0) = c_1 + c_2$$

$$y(1) = -c_1 + 4c_2$$

By equating these two sets of relations, we have

$$c_1 + c_2 = 3y(-1) + 4y(-2) \qquad \therefore c_1 = -\frac{1}{5}y(-1) + \frac{4}{5}y(-2)$$

$$-c_1 + 4c_2 = 13y(-1) + 12y(-2) \qquad c_2 = \frac{16}{5}y(-1) + \frac{16}{5}y(-2)$$

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Therefore, the zero-input response of the system is

$$y_{zi}(n) = \left[-\frac{1}{5}y(-1) + \frac{4}{5}y(-2)\right](-1)^n + \left[\frac{16}{5}y(-1) + \frac{16}{5}y(-2)\right]4^n \qquad n \geq 0$$

For example, if $y(-2)=0$ and $y(-1)=5$ then $C_1=-1$, $C_2=16$

Hence,

$$y_{zi}(n) = (-1)^{n+1} + (4)^{n+2} \qquad n \geq 0$$

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Linear Constant Coefficient Difference Equation (LCCDE)

Solution has two parts
- → Homogeneous solution/Natural/ zero input solution ($y_h[n]$)
- → Final state/steady state solution ($y_s[n]$)

# Example 1:

Solve $y[n] - 3y[n-1] - 4y[n-2] = x[n] + 2x[n-1]$ for $x[n] = \delta[n]$. Consider the system is initially relaxed.
↳ Initial conditions = 0

Solution:

For impulse input, steady state response $\triangleq y_s[n] = 0$

Now, for determining zero input solution,

$$\lambda^n - 3\lambda^{n-1} - 4\lambda^{n-2} = 0$$

$$\Rightarrow \lambda^n\left(1 - \frac{3}{\lambda} - \frac{4}{\lambda^2}\right) = 0$$

$$\Rightarrow \lambda^2 - 3\lambda - 4 = 0 \quad \left[\text{Multiplying both sides by } \frac{\lambda^2}{\lambda^n}\right]$$

$$\Rightarrow \lambda^2 - 4\lambda + \lambda - 4 = 0$$

$$\Rightarrow \lambda(\lambda - 4) + 1(\lambda - 4) = 0$$

$$\Rightarrow (\lambda - 4)(\lambda + 1) = 0$$

$$\therefore \lambda = 4, -1$$

$$\therefore y_h[n] = C_1(4)^n + C_2(-1)^n$$

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$\therefore\ y[n] = C_1 4^n + C_2(-1)^n \quad \cdots (I)$

Given,

$$y[n] = x[n] + 2x[n-1] + 3y[n-1] + 4y[n-2] \quad \cdots (II)$$

From (I),

$$y[0] = C_1 + C_2$$

From (II), $y[0] = x[0] + 2x[-1] + 3y[-1] + 4y[-2]$

$$= \delta[0] + 2\delta[-1] + 3y[-1] + 4y[-2]$$

$\therefore y[0] = 1$ [$\because \delta[-1] = 0$; since system is initially relaxed, $y[k] = 0$ for $k<0$]

$\therefore\ C_1 + C_2 = 1 \quad \cdots (III)$

Again from (I),

$$y[1] = 4C_1 - C_2$$

From (II),

$$y[1] = \delta[1] + 2\delta[0] + 3y[0] + 4y[-1]$$

$$= 2\times1 + 3\times1 = 5$$

$\therefore\ 4C_1 - C_2 = 5 \quad \cdots (IV)$

Solving (III) and (IV), $C_1 = \frac{6}{5}$ ; $C_2 = -\frac{1}{5}$

$\therefore y[n] = \frac{6}{5}(4)^n - \frac{1}{5}(-1)^n$ (Ans)

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# Example 2

Solve, $y[n] - 3y[n-1] - 4y[n-2] = x[n] + 2x[n-2]$ for unit step input $\{x(t) = u(t)\}$.

Initial conditions : $y[-1] = 3$
$y[-2] = 2$

Same problem but different input and initial conditions.

Solution:

For homogeneous or zero input solution:

$$\lambda^n - 3\lambda^{n-1} - 4\lambda^{n-2} = 0$$

$$\Rightarrow \lambda = 4, -1 \quad \text{[Same as before]}$$

$$\therefore y_h[n] = C_1(4)^n + C_2(-1)^n \quad \ldots (1)$$

For steady state solution:

Since $x[n] = u[n]$; $y(n) = k\,u(n)$

$$\therefore K u(n) - 3k\,u[n-1] - 4k\,u[n-2] = u[n] + 2u[n-2]$$

At steady state, $K - 3k - 4k = 1 + 2$

$$\Rightarrow k = \frac{3}{-6} = -\frac{1}{2}$$

$$\therefore y_s[n] = -\frac{1}{2}u[n]$$

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$$\therefore y[n] = C_1(4)^n + C_2(-1)^n - \frac{1}{2}u[n] \quad \ldots (I)$$

Given,

$$y[n] = u[n] + 2u[n-1] + 3y[n-1] + 4y[n-2] \ldots (II) \quad [\because x[n] = u[n]]$$

Putting $n=0$ in (I) and (II),

$$y[0] = C_1 + C_2 - \frac{1}{2}$$

and $y[0] = u[0] + 2u[-1] + 3y[-1] + 4y[-2]$

$$= 1 + 2\times 0 + 3\times 3 + 4\times 2$$

$$\therefore y[0] = 18$$

$$\therefore C_1 + C_2 - \frac{1}{2} = 18$$

$$\Rightarrow C_1 + C_2 = \frac{37}{2} \quad \ldots (III)$$

Again, putting $n=1$,

$$y[1] = 4C_1 - C_2 - \frac{1}{2}$$

and $y[1] = u[1] + 2u[0] + 3y[0] + 4y[-1]$

$$= 1 + 2\times 1 + 3\times 18 + 4\times 3$$

$$y[1] = 69$$

$$\therefore 4C_1 - C_2 - \frac{1}{2} = 69$$

$$\Rightarrow 4C_1 - C_2 = \frac{139}{2} \quad \ldots (IV)$$

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Solving (iii) and (iv),

$$C_1 = \frac{88}{5} \quad ; \quad C_2 = \frac{9}{10}$$

$$\therefore y[n] = \frac{88}{5}(4)^n + \frac{9}{10}(-1)^n - \frac{1}{2}u[n] \quad \text{(Ans)}$$

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